

# DEXTER FERNANDES

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## SUMMARY

Robotics software engineer with an MSc in Robotics (Merit) and 3+ years of commercial C++ and computer vision engineering experience. Built a real-time C++17 and ROS2 LiDAR SLAM system on a Clearpath Husky, covering localisation, odometry, sensor fusion and sensor calibration, with solver backends in Ceres, g2o and GTSAM. Delivered production C++ inference pipelines with gtest coverage, CI and dependency management via Boost and CMake, and comfortable turning research and prototype code into tested, maintainable systems. Works independently across complex, cross-team codebases, from industrial edge deployments to on-robot ROS2 stacks, documenting solutions for handover.

## RELEVANT EXPERIENCE

### Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Hardened deployed DeepStream and FFmpeg video pipelines with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live pipeline crashes by 67% and enabling recovery without site visits.
- Built a watchdog application with a REST API exposing service health and restart control, so operators and other services could act on failures remotely without an engineer on site.
- Introduced pytest, unittest and linting into CI for edge inference modules, improving release consistency across hardware targets.
- Added observability with structured logs, metrics, timeouts, retries and message validation, cutting root-cause analysis time on live deployments.
- Owned the end-to-end ML lifecycle for real-time perception on BP rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Built ingestion and validation pipelines across image, video and sensor data with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout from 72h to under 24h.

### LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, overlapping preprocessing, inference and post-processing to raise throughput from 5 to 15 FPS under real-time constraints.
- Wrote gtest unit tests and documented interfaces for the C++ inference path, and integrated Boost C++ libraries into production pipelines, resolving dependency and build issues to keep deployments reproducible.
- Profiled C++ inference and pre/post-processing hotspots with Intel VTune, directing architecture fixes in the CNN inference pipeline.
- Led four engineers building a two-stage C++ detector pipeline for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.
- Owned production ML pipelines end to end, from ingestion through training, optimisation, deployment and field validation for industrial safety systems.

## PROJECTS

### DeepStream RTSP Pipeline [github.com/Dexter-Fernandes/deepstream-rtsp-pipeline](https://github.com/Dexter-Fernandes/deepstream-rtsp-pipeline)

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction and re-streamed output.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Built an automated model-promotion gate checking match-rate and mean-IoU against a signed SHA-256 manifest, and wired GPU smoke plus MOT17 integration tests into GitHub Actions CI.
- Hardened cross-camera fusion for real deployments: derived NTP-based timestamps across RTSP sources, gated fusion on clock skew, and incremented reconnect generations to stop stale tracker-ID reuse.

### Modern Edge CV Benchmarking [github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking](https://github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking)

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to assess deployment suitability for edge and embedded targets.
- Exported models to ONNX and evaluated OpenVINO deployment paths, comparing inference time and model size to support library and framework selection decisions.

### ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer with robot\_localization fusing wheel odometry and IMU for state estimation.
- Developed ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2 and message synchronisation to debug transform timing issues.

## EDUCATION

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (rclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Configured Ceres, g2o and GTSAM as solver backends, tuned ICP parameters, and calibrated LiDAR extrinsics to the robot base, resolving transform timing with tf2 across a PCL point cloud pipeline.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier under JetPack, working within the platform's compute and memory limits for on-robot operation.

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

|                          |                                                                                                                                                                                                                                                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages                | C++17, CUDA C++, Python, Bash, SQL                                                                                                                                                                                                                                                                                  |
| Robotics and SLAM        | ROS2 (Robot Operating System 2), rclcpp, tf2, rosbag2, Gazebo, Nav2, LiDAR SLAM, localisation, odometry, sensor fusion, state estimation, pose graph optimisation, ICP (Iterative Closest Point), LiDAR extrinsics calibration, PCL (Point Cloud Library), Open3D, robot_localization, EKF (Extended Kalman Filter) |
| Optimisation and Solvers | Ceres, g2o, GTSAM, Eigen, Bayesian optimisation                                                                                                                                                                                                                                                                     |
| Architecture and APIs    | modular pipeline architecture, REST API design, containerised microservices, dependency management, CMake, Boost                                                                                                                                                                                                    |
| Testing and CI/CD        | gtest, pytest, unittest, GitHub Actions, CI/CD, Git, integration testing                                                                                                                                                                                                                                            |
| Production and Edge      | Docker, NVIDIA Jetson AGX Xavier, JetPack, Intel Movidius VPU, TensorRT (custom plugins), OpenVINO, structured logging, health monitoring, observability, Linux                                                                                                                                                     |
| Video and Perception     | NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, multi-camera synchronisation, OpenCV                                                                                                                                                                                                                                    |
| Computer Vision and ML   | PyTorch, YOLO, object detection, object tracking, quantisation                                                                                                                                                                                                                                                      |
| Professional             | technical leadership, mentoring, Agile and Scrum, stakeholder communication, cross-functional delivery                                                                                                                                                                                                              |

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering

05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

Taco Bell — Service Champion / Team Trainer

09/2024 – PRESENT · BRISTOL, ENGLAND

Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)

Best Speaker award, ABC Toastmasters Club (01/2020)